

Homework 3

Math 2031 — Professor Hamilton

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Section 5.2

Question 1C

Statement: $1^3 + 2^3 + 3^3 + \dots + n^3 = \frac{n^2(n+1)^2}{4}, n \in \mathbb{N}$

Proof by Induction:

Base Case: $k = 1, k \in \mathbb{N}$

LHS: $1^3 = 1$

RHS: $\frac{1^2(1+1)^2}{4} = \frac{4}{4} = 1$

Since LHS = RHS, the base case is true.

Induction Step:

Assume $1^3 + 2^3 + 3^3 + \dots + k^3 = \frac{k^2(k+1)^2}{4}$ is true for k .

We want to show the statement is true also for $(k+1)$ by showing the LHS=RHS.

Substituting $(k+1)$ into the equation:

$$1^3 + 2^3 + 3^3 + \dots + k^3 + (k+1)^3 = \frac{(k+1)^2(k+2)^2}{4}$$

Here we notice our IH:

$$\underbrace{1^3 + 2^3 + 3^3 + \dots + k^3}_{\text{Inductive Hypothesis}} + (k+1)^3 = \frac{(k+1)^2(k+2)^2}{4}$$

Replace:

$$\frac{k^2(k+1)^2}{4} + (k+1)^3 = \frac{(k+1)^2(k+2)^2}{4}$$

Factor $(k+1)^2$ from the LHS:

$$(k+1)^2 \left(\frac{k^2}{4} + k + 1 \right) = \frac{(k+1)^2(k+2)^2}{4}$$

Divide both sides by $(k+1)^2$, which is greater than 0.

$$\frac{k^2}{4} + k + 1 = \frac{(k+2)^2}{4}$$

Expanding the LHS we get:

$$\frac{k^2}{4} + k + 1 = \frac{k^2}{4} + k + 1$$

Thus $\forall n \in \mathbb{N}, 1^3 + 2^3 + 3^3 + \dots + n^3 = \frac{n^2(n+1)^2}{4}$

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Question 2A

Statement: $1^2 + 3^2 + 5^2 + \dots + (2n-1)^2 = \frac{(2n-1)(2n)(2n+1)}{6}, n \in \mathbb{N}$

Proof by Induction:

Base Case: $k = 1, k \in \mathbb{N}$

$$\text{LHS: } (2(1) - 1)^2 = 1^2 = 1$$

$$\text{RHS: } \frac{(2(1)-1)(2(1))(2(1)+1)}{6} = \frac{1 \cdot 2 \cdot 3}{6} = \frac{6}{6} = 1$$

Since LHS = RHS, the base case is true.

Induction Step:

Assume $1^2 + 3^2 + 5^2 + \dots + (2k-1)^2 = \frac{(2k-1)(2k)(2k+1)}{6}$ is true for k .

We want to show it is also true for $k+1$ by showing that the LHS=RHS.

Substituting $k+1$ into our equation:

$$1^2 + 3^2 + 5^2 + \dots + (2k-1)^2 + (2k+1)^2 = \frac{(2k+1)(2k+2)(2k+3)}{6} \text{ is true for } k.$$

Note our IH:

$$\underbrace{1^2 + 3^2 + 5^2 + \dots + (2k-1)^2}_{\text{Inductive Hypothesis}} + (2k+1)^2 = \frac{(2k+1)(2k+2)(2k+3)}{6}.$$

Substitute:

$$\frac{(2k-1)(2k)(2k+1)}{6} + (2k+1)^2 = \frac{(2k+1)(2k+2)(2k+3)}{6}$$

Factoring a $(2k+1)$ on both sides:

$$(2k+1) \cdot \left(\frac{(2k-1)(2k)}{6} + (2k+1) \right) = (2k+1) \cdot \frac{(2k+2)(2k+3)}{6}$$

Canceling the $(2k+1)$, it is greater than 0.

$$\frac{(2k-1)(2k)}{6} + (2k+1) = \frac{(2k+2)(2k+3)}{6}$$

Expanding both sides:

$$\frac{4k^2+10k+6}{6} = \frac{4k^2+10k+6}{6}$$

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$$\text{Thus } \forall n \in \mathbb{N}, 1^2 + 3^2 + 5^2 + \dots + (2n-1)^2 = \frac{(2n-1)(2n)(2n+1)}{6}$$

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Question 3

Statement: For any real number a except 1, $1 + a + a^2 + a^3 + \dots + a^n = \frac{(a^{n+1}-1)}{a-1}$